

RESEARCH VISION

My research is centered on developing advanced machine learning methodologies for reliable and decision-relevant artificial intelligence with a particular emphasis on integrating domain knowledge and mathematical optimization into learning, improving model generalizability under limited and heterogeneous data, and translating methodological advances into practical decision-support systems. I pursue these challenges across critical engineering and scientific domains, including healthcare, intelligent manufacturing, and smart agriculture (see Figure 1). During my Ph.D., I have collaborated with four interdisciplinary research groups spanning industrial and systems engineering, sustainable bioproducts, radiation oncology, and biomedical research, and contributed to six research projects, including externally sponsored projects supported by USDA–NIFA, NIH, and the Mississippi Institutions of Higher Learning. As a mentor, I have co-mentored two graduate and five undergraduate researchers in machine learning and AI research, contributing to peer-reviewed publications, posters, and research presentations.

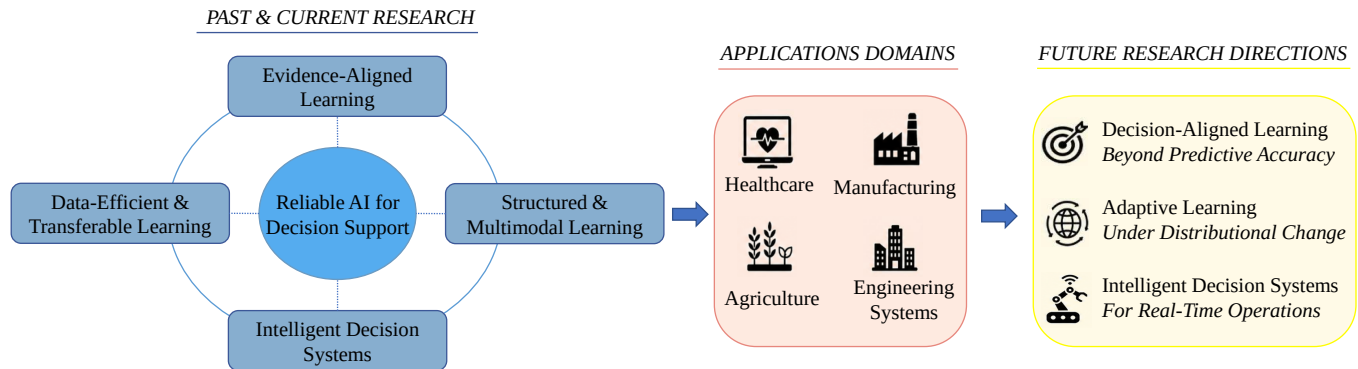


Figure 1: An overview of my research roadmap.

Machine learning models are increasingly being used to support decisions in complex scientific and engineering systems, yet several fundamental challenges limit their reliability in real-world settings. First, strong predictive performance does not necessarily guarantee that a model’s decision is supported by the right evidence. For example, an object detector may accurately identify many individual objects while still distorting the population-level distribution needed for manufacturing quality assessment, while a medical imaging model may produce a correct prediction based on anatomically irrelevant regions. Second, labeled data are often limited, costly to obtain, or heterogeneous across environments, requiring models that can learn efficiently and remain effective under domain shifts. Third, many real-world decisions depend on structured and multimodal information, including images, physiological signals, and environmental or process sensors, whose complementary relationships must be represented and integrated effectively. Finally, even an accurate and reliable model has limited practical value if it cannot operate efficiently within the computational, latency, and usability constraints of real decision environments. My research addresses these interconnected challenges through four complementary directions: evidence-aligned and constrained learning, data-efficient and transferable learning, structured and multimodal machine learning, and the translation of learning methodologies into intelligent decision-support systems.

PAST & CURRENT RESEARCH CONTRIBUTIONS

1. Evidence-Aligned and Constrained Learning

Predictive models can achieve strong performance while still relying on incomplete or misleading evidence, limiting their reliability for real-world decisions. My research addresses this problem by developing evidence-aligned constrained learning methods that incorporate domain-relevant information directly into model training.

In collaboration with the Department of Sustainable Bioproducts, I developed *UOT-DETR*, which integrates unbalanced optimal transport with visual detection to preserve population-level size distributions in dense industrial inspection. This formulation reframes dense object detection as a population-level quality-measurement problem rather than solely an instance-level localization task. The approach improves the recovery of small and occluded objects while producing more representative physical measurements, enabling more reliable quality monitoring and process decision-making in biomass manufacturing.

I extended this direction through collaboration with radiation oncology researchers at the University of Mississippi Medical Center, where I developed a *knowledge-informed constrained learning* framework for preoperative assessment of extranodal extension in head-and-neck cancer. By embedding anatomical priors into feature learning and decision-making, the approach guides the model toward clinically meaningful regions, reduces the effective learning space, and improves interpretability and data efficiency under limited supervision. Together, these studies establish a general framework for aligning model-induced evidence with the physical or anatomical evidence required by the decision problem.

2. Data-Efficient and Transferable Learning

Limited supervision and distribution shift remain major barriers to reliable machine learning, particularly in medical imaging where expert annotations are costly and data vary across institutions. To address label scarcity, I integrated *self-supervised representation learning* with my knowledge-informed framework, using masked image modeling to learn domain-specific anatomical representations from unlabeled CT data before supervised fine-tuning. In complementary work, I contributed to *DA-GMGNet*, a domain-adaptation framework for improving 3D CT-based cancer assessment across heterogeneous data distributions. Beyond representation transfer, I investigated knowledge transferability in *differentiable Neural Architecture Search (NAS)*, examining when architectural knowledge discovered for one medical-imaging task can be reused in another. Together, these studies address transfer at three complementary levels: representation, domain, and architecture, improving data efficiency, cross-domain robustness, and the reuse of learned knowledge across related problems.

3. Learning from Structured and Multimodal Data

Complex decision problems often involve heterogeneous signals whose relationships carry more information than individual measurements alone. My research investigates graph-based and multimodal learning to capture these interactions and transform multi-source data into decision-relevant representations.

In collaboration with researchers at the University of Mississippi Medical Center Pediatric Sleep Center, I developed a graph-theoretic machine-learning framework for ADHD assessment using multichannel polysomnography (PSG) data, including EEG, EOG, EMG, and ECG signals. By modeling functional relationships among physiological signals across sleep stages, the approach identified connectivity-based biomarkers that complement conventional aggregate sleep measures.

I extended this direction to smart agriculture through collaboration with the Department of Biochemistry, Nutrition, and Health Promotion, where I co-developed a multimodal IoT platform that integrates heterogeneous environmental and plant-health sensor streams with time-series analytics and machine-learning-based prediction. Together, these studies advance my broader interest in learning from relational, multimodal, and temporally evolving data for intelligent decision support.

4. Translating Machine Learning into Intelligent Decision-Support Systems

Methodological advances have limited impact if they remain disconnected from the systems in which decisions are actually made. A complementary component of my research focuses on bridging the gap between machine-learning methodology and practical intelligent systems. I am particularly interested in how sensing, learning, analytics, and human decision-making can be integrated into complete end-to-end frameworks.

In intelligent manufacturing, I translated my visual quality-monitoring research into a portable edge-AI system capable of continuous on-device inspection, quality assessment, alert generation, and interactive visualization. In smart agriculture, I co-developed a multisensor monitoring platform that integrates field data acquisition, predictive analytics, historical analysis, and configurable decision support. I have also developed an engineering risk-analysis platform that combines multi-hazard modeling, economic analysis, and geospatial information to support risk-informed decisions. Together, these efforts define a systems-oriented research direction in which sensing, learning, analytics, and decision-making are integrated to deliver reliable, deployable decision support in real-world engineering environments.

FUTURE RESEARCH DIRECTIONS

The focus of my lab will be to advance reliable and decision-faithful AI for complex engineering and scientific systems. My research will address three fundamental challenges: aligning learning objectives with real decision requirements,

maintaining model effectiveness as data distributions evolve, and integrating AI into real-time physical and operational decision systems. These challenges define the following three research directions.

1. Decision-Aligned Learning Beyond Predictive Accuracy

Most machine-learning models are trained to minimize prediction error, although in many engineering and scientific applications the true objective is the quality of the decision that follows. My lab will investigate optimization-integrated learning frameworks that incorporate decision-relevant evidence, risk, and operational objectives directly into model training. A central direction will be the development of adaptive constraints that respond to violations during learning. This formulation will allow different aspects of model behavior to be prioritized according to their importance to the decision problem. A key methodological question will be how to balance predictive accuracy, evidential validity, and operational risk when these objectives cannot be simultaneously optimized. The goal is to move from prediction-centered models toward learning systems whose objectives reflect the physical, clinical, or operational consequences of the decisions they support.

2. Adaptive Learning Under Distributional Change

Machine-learning models are commonly developed under fixed training conditions, although the environments in which they operate rarely remain stationary. Changes in patient populations, sensing technologies, acquisition protocols, or operating conditions can alter data distributions and degrade model performance. My lab will investigate learning systems that can identify and adapt to evolving data distributions rather than assuming that training and deployment conditions remain unchanged. A major focus will be on developing models that preserve useful knowledge while adapting to new environments without requiring complete retraining or extensive new annotations. A key research question will be how to balance adaptation to new environments with the retention of previously learned knowledge as operating conditions continue to evolve. Such capabilities are essential for machine-learning systems expected to operate reliably across institutions, production settings, sensing platforms, and other changing real-world environments.

3. Intelligent Decision Systems for Real-Time Operations

Machine-learning models are often developed as standalone predictors, although real engineering systems require predictions to be translated into timely and actionable decisions. In manufacturing, agriculture, and healthcare, intelligent systems must integrate sensing, computation, and decision-making while operating under constraints on time, resources, and available information. My lab will focus on developing end-to-end intelligent decision systems that determine what information should be collected, when the available information is sufficient to support an action, and how feedback from previous decisions can improve subsequent behavior. A key research question will be how to coordinate sensing, computation, and decision-making under resource constraints so that limited information and computational capacity are used where they provide the greatest value to the resulting action. A key direction will be to integrate AI with edge and robotic systems so that sensing, inference, and decision-making can occur close to the physical process. These capabilities can enable more responsive, autonomous, and human-in-the-loop decision systems for complex real-world operations.

POTENTIAL FUNDING SOURCES

Since my future research goals are centered on developing and advancing methodologies for reliable AI, optimization, and intelligent decision systems, I will pursue funding support from the National Science Foundation (NSF), particularly through programs such as Robust Intelligence (RI), Operations Engineering (OE), Cyber-Physical System Foundations (CPS), and Mathematical Foundations of Artificial Intelligence (MFAI). I will also seek collaborations with medical and clinical research teams and agriculture-related departments to develop application-driven proposals for the National Institutes of Health (NIH) and the U.S. Department of Agriculture (USDA), respectively. More broadly, I plan to collaborate within and outside the department, including across research centers and external institutions, to pursue interdisciplinary internal and external funding opportunities.